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## Development of an Influencer-Brand Congruence Measurement Framework for Recommender Systems

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### **Abstract:**

Development of celebrity-brand, and subsequently influencer-brand congruence measurement scale has been the main aim of many studies since the 1950s. Currently, there is a growing interest in identifying the prerequisites of a successful influencer brand endorsement or influencer brand collaboration in order for the brands to determine which creators represent the most suitable match. While previous studies have addressed this challenge with the help of credibility or attitude homophily measures, this article presents a new perspective by combining existing findings with the recommender system mechanisms. Following the principles of collaborative and content-based filtering, we implement binary Jaccard similarity to calculate the congruence score between a brand and a content creator based on the prerequisite of the Kamins match-up hypothesis.

**Keywords:** influencer-brand congruence, influencer marketing, Jaccard similarity, match-up hypothesis, product endorsement

### **INTRODUCTION**

Brand personality refers to the set of human characteristics associated with a brand creating a unique and recognizable identity beyond the brand's functional features and benefits (Aaker, 1997; Donvito et al., 2020). Since the 1950's brands have been looking for strategies on how to enhance this personality and achieve a more significant connection with their potential customers. Celebrity endorsement has been considered to be one of the most effective approaches thanks to the availability to extend basic brand-consumer relationships by providing the para-social aspect (Halder et al., 2021). Nowadays, this concept has evolved building upon even stronger para-social relationships (Jones et al., 2022). Influencer endorsement is a marketing strategy in which the influencer uses their personal brand to endorse the brand by creating content such as posts, stories, or videos that showcase their experiences with the product or service in front of their followers. The idea behind this strategy is that the audience's identification with the creator will be significant enough to transfer the endorsed brand or product (Belanche et al., 2021). There has been a growing body of research devoted to the antecedents of influencer endorsement effectiveness. While the number of influential factors and the connection between them differ, the majority of authors agree that influencer-brand-product fit is a key prerequisite (Halder et al., 2021).

Being aware of the growing potential of influencer brand collaborations, many new influencer brand platforms have been emerging. Influencer brand platforms are websites that provide a centralized space for brands to discover new creators suitable for cooperation, and for influencers to showcase their personal brand and content creators skills. Popular influencer brand platforms include for example: Influencer.co, AspireIQ, Tribe, Upfluence or NeoReach

(Geyser, 2022). New tools and platforms for brands and creators are being developed also by social networking sites. Meta recently introduced the Brand Collabs Manager (Newberry, 2022) and YouTube Launched the Brand Connect (Scott-Green, 2020). These platforms often offer features such as influencer discovery, campaign management, analytics and reporting, and payment processing. Influencer brand platforms are designed to help brands and influencers work together effectively to reach their shared objectives. It is challenging to determine the exact amount of money being invested in influencer marketing in general and in influencer brand platforms specifically since this information is not publicly available for many companies. However, according to Geyser (2023), some of the largest influencer brand platforms have received substantial investments from venture capital firms and other investors. For example, AspireIQ raised \$18 million in a Series B funding round in 2019, while NeoReach raised \$6 million in a Series A funding round in 2020.

Investment in influencer brand platforms is driven by the growth of the influencer marketing industry, as well as the increasing recognition of the value of connecting brands with influencers. Brands can use these platforms to find influencers who match their target audience and goals, and to streamline the influencer marketing process (Santora, 2023). However, matching brands with influencers only based on similar target groups is not sufficient. Many studies have already demonstrated that influencer brand congruence is a complex concept corresponding to the degree to which an influencer's personal brand aligns with the brand they are endorsing (Halder et al., 2021; Donvito et al., 2020; Zu et al., 2019; Kamins, 1990). Although the emerging platforms provide a valuable connection for both brands and influencers, they represent only the first step in measuring the level of congruence between the collaborators. Developing an easy to implement framework could help both parties to assess the potential benefits and risks of such collaboration.

The aim of this paper is to propose an influencer brand congruence framework suitable for both marketers and recommender systems used by the emerging collaborator platforms. Based on the four most widely used approaches to celebrity brand congruence (Source credibility model (Hovland and Weiss, 1951), Source attractiveness model (McGuire's, 1985), Meaning transfer model (McCracken, 1989), and the match-up hypothesis (Kamins, 1990) we establish the defining characteristic of influencer brand congruence. Subsequently, we combine these findings with rating mechanisms implemented by recommender systems to reflect individual preferences of the users. In this respect, we treat brands as “users” with specific marketing needs searching for tailored content provided by influencers. Finally, we propose a measuring scale as an integral part of the influencer-brand matching mechanism.

## **1 The Role of Perceived Similarity**

The relationship between potential customers and figures they follow on social media has been viewed as an increasingly essential aspect in influencing the consumer's attitude toward the celebrity-endorsed brand and in predicting endorsement efficacy (Zhang et al., 2021). Influencers' ability to regularly interact with their audiences through engaging social media exchanges represents a crucial prerequisite for establishing parasocial interactions (PSI) between the online creators and their followers. The theory of parasocial interactions, firstly introduced by Horton & Wohl (1956), is a psychological concept originally related to traditional media. According to Giles (2009), once an individual made his or her personal judgment about a media figure or assigned them personal traits, they will start perceiving the figure as being part of their physical space and thereby allowing the media figure to become part of their social network. While the development and strength of PSI is conditioned by many factors, homophily seems to be a recurring predictor in both traditional media studies

and recent research regarding influencer marketing (Ladhari et al., 2020; Sokolova & Kefi, 2020; Levine & Valle, 1975; McCroskey et al., 1974; Brock, 1965).

In 1965, Brock described similarity between the communicator and the recipient as a factor facilitating persuasion. According to Levine & Valle (1975), when the audience is able to identify itself with the speaker, the audience perceives the speaker as more trustworthy. Similarly, social cognitive theory implies that individuals are more likely to follow and adopt the opinions of personas with which they share similar characteristics (McCroskey et al., 1974).

Skolova & Kefi (2020) analyzed the responsiveness of consumers to Instagram and YouTube influencers' endorsed products. Their findings indicate that attitude homophily is significantly connected to both credibility and PSI that influence the final purchase intention. Ladhari et al., 2020 focused on four dimensions of homophily including attitude, value, background and appearance. They came to the conclusion that three out of the four dimensions (background excluded) have an impact on the YouTuber's popularity. The findings are consistent with Ohanian (1990) who puts emphasis on three crucial qualities: physical appeal, knowledge, and reliability. Evidence generally implies that a message's persuasiveness is correlated with how much its audience likes the message's information source (Balabanis & Chatzopoulou, 2019). According to Pornpitakpan's (2004), each of the three source traits has a different weighting, with trustworthiness being more significant than both knowledge and physical appeal.

In their study, Lazarsfeld and Merton (1954) found that human beings tend to form relationships with others who share similar attitudes and beliefs, and that these relationships are important in shaping their attitudes and beliefs. They also found that people are more likely to be influenced by the opinions of their friends and family members, rather than by the opinions of public figures or the media. They proposed that attitude homophily occurs due to a combination of two factors: selection and socialization. Selection refers to the idea that people tend to seek out and associate with others who have similar attitudes and beliefs. This can occur through processes like homophilous friendship formation, in which people tend to form friendships with others who have similar attitudes. Socialization refers to the idea that people's attitudes and beliefs are influenced by their interactions with others. According to Lazarsfeld and Merton (1954), people tend to adopt the attitudes and beliefs of those with whom they associate, which can lead to increased homophily over time.

In general, the majority of authors agree that perceived similarity is a significant factor in the formation of para-social relationships, identification with the influencers and through them also with the brand. However, the existing studies differ on the role of the perceived similarity. While some authors introduce similarity as an direct antecedent of the relationship formation between influencers and their followers (Ladhari et al., 2020; Sokolova & Kefi, 2020; Levine & Valle, 1975; Lazarsfeld and Merton, 1954) others believe that similarity has a moderating effect on trustworthiness (Levine & Valle, 1975; Hovland and Weiss, 1951), attractiveness (McGuire's, 1885) and cultural, social and meaning transfer (McCraken, 1989).

## **2 The Celebrity Endorsement Models**

The perceived similarity appears to be important not only in the relationship between influencers and followers but also between influencers and brands they endorse. Influencer brand congruence refers to the alignment or consistency between the values, image and message of an influencer and the brand they are promoting (Kamins, 1990). When there is a

high level of congruence, the influencer's followers are more likely to view the brand positively and be more likely to purchase the products or services being promoted. This is why it is important for brands to choose influencers whose personal brand aligns with their own. According to Kahle and Homer (1985) and Kamins (1990), the perceived fit between the celebrity's image and the promoted brand determines how persuasive an advertisement will be. In addition, Kamins (1990) and Fleck, Korchia, & Le Roy (2012) suggest that this perceived compatibility extends physical attractiveness.

Identifying the factors that have impact on the congruence between the celebrity and the brand and the consequent effectiveness of the celebrity endorsement has been the goal of many models since the 1950s. Torres et al. (2019) divide the existing models in three main categories: attractiveness-based, expertise-based, and image-based. All three categories require achieving a certain degree of homophily between the consumer, influencer and endorsed product. The most notorious models include the Source credibility model proposed by Hovland and Weiss (1951), McGuire's (1985) Source attractiveness model, the meaning transfer model (McCracken, 1989), and the match-up hypothesis employed by Kamins (1990).

The source credibility model is a communication-based model that explains how the credibility of a message source affects the persuasiveness of a message. Introduced by psychologists Hovland and Weiss (1951), the model proposes that the credibility of a source is composed of two main factors: expertise and trustworthiness. Expertise refers to the source's perceived knowledge and skill in the topic being discussed. Trustworthiness refers to the source's perceived honesty, integrity and reliability. The model suggests that a message from a source that is perceived as both expert and trustworthy is more likely to be accepted and acted upon than a message from a source that is perceived as lacking in one or both of these characteristics. Consistently with Hovland et al. (1949), Hovland and Weiss (1951) demonstrated that while source credibility has no impact on information acquisition and retention, it is able to induce opinion shifts.

The McGuire's (1985) model of source attractiveness builds upon the credibility model and extends the original model by adding a third factor: similarity. According to this theory, a source will be evaluated as more attractive if they are perceived as knowledgeable and skilled (expertise), honest and reliable (trustworthiness), and similar to the observer in terms of attitudes, values, and other characteristics (similarity). Influencer source attractiveness can be influenced by factors such as the influencer's level of expertise in their field, their level of engagement with their audience, the authenticity of their content, previous brand collaborations, audience demographics, and other metrics. Influencers with high source attractiveness are typically seen as more credible and are therefore more effective at promoting products or services to their audience.

The meaning transfer model developed by McCracken in 1989 states that linking a celebrity to a product or a brand might alter its perception. According to the meaning transfer theory, when consumers are exposed to sponsored content through celebrity endorsement a process that converts the meaning of a celebrity to a brand takes place. During this process, the values associated with the influencer's personality are transmitted to the brand (Torres et al., 2019). Torres et al. (2019) believe the meaning transfer model to be a more accurate brand-celebrity congruence representation than the source attractiveness and source credibility models. These models fail to capture the potential influence of the celebrities personal characteristics and likeability on the endorsement process.

The meaning transfer model suggests that consumers use products and brands to give meaning to their lives and express their identities. According to the model, consumers transfer the meanings and symbols associated with a product or brand to themselves, using it to create a sense of self and belonging. The model proposes three main types of meaning transfer:

1. Cultural meaning transfer, in which consumers adopt the values and beliefs associated with a product or brand as their own.
2. Social meaning transfer, in which consumers use a product or brand to signal their membership in a particular group or social category.
3. Personal meaning transfer, in which consumers use a product or brand to express their own unique identity and personality.

Similarly, Kamins in the match-up hypothesis (1990) proposes that the effectiveness of a source of information depends on how well the characteristics of the source match the characteristics of the message. According to the theory, a source that is perceived as similar to the target audience in terms of demographic characteristics, attitudes, or values will be more effective in delivering a message than a source that is perceived as dissimilar. While being frequently used in celebrity endorsement research (Schimmelpfennig and Hunt, 2019; Törn, 2012), some authors criticize the match-up hypothesis for neglecting the asymmetry of the endorsement effects (Ang and Dubelaar, 2006). To reflect the asymmetry, Ang and Dubelaar (2006) proposed an extended version of the match-up hypothesis called co-activation theory. This theory builds upon the perceived similarity, however rather than measuring common characteristics between the celebrity source and endorsed product, focuses on the associations that are co-activated when a celebrity and brand are juxtaposed together. The authors distinguish supporting and opposing activations. Every endorsement can trigger both and the brand-product-celebrity fit is established based on the difference between the two counteracting categories of associations.

Analogically to the influencer-follower relationship, perceived similarity can be linked to all of the above described models either as the main component or as a prerequisite for influencer brand endorsement effectiveness. Since similarity represents a recurring element in the majority of studies on PSI between influencer and followers and also in influencer brand endorsements, we believe that this perceived similarity can be used as a quantitative measure that would allow recommender systems to match the creators with suitable brand counterparts.

### **3 Matching Mechanisms in Recommender Systems**

A recommender system, also known as a recommendation system, is a subclass of information filtering system that seeks to predict the "rating" or "preference" a user would give to an item (Iovine et al., 2020). Recommender systems are utilized in a variety of areas such as playlist generators for video and music services, product recommenders for online stores, or content recommenders for social media platforms (Ghasemi and Momtazi, 2021). There are several algorithms used to implement recommender systems, including:

- Collaborative Filtering: This approach makes recommendations based on the knowledge of users' attitude to items, that is, it uses the "wisdom of the crowd" to recommend items (Behera and Nain, 2023).
- Content-Based Filtering: This approach uses item metadata, such as genre, director, actor, etc., to make recommendations. The basic idea is that if a person liked a

particular item, he or she will also like an item that is similar to it (Son and Kim, 2017).

- Hybrid Recommender Systems: This approach combines both collaborative filtering and content-based filtering to make recommendations (Walek and Fajmon, 2023).

In all three cases, the algorithm first needs to represent each item as a set of features or attributes. Next, the algorithm builds a user profile based on their preferences. To make recommendations, the algorithm calculates the similarity between the user's profile and the features of each item in the database. The items that are most similar to the user's profile are then recommended to the user (Afoudi et al., 2021). The similarity between two items or a user's profile and an item can be calculated using various similarity measures. According to Liu et al. (2017), similarity measures can be divided into four main categories: the predefined similarities, the learned similarities, the preference-based similarities and the mixed similarities. These similarity measures provide a way to quantify the similarity between two items or a user's profile and an item. The choice of similarity measure depends on the type of data and the goals of the recommender system.

In the context of influencer-brand recommender platforms, brands are perceived as users and influencer profiles as items. In addition, the recommendation should be done not based on the brand's preferences, but based on the degree of similarity between the brand's profile and influencer's profile. Therefore content-based filtering appears to be a more suitable recommender approach (Kotu and Deshpande, 2019). Content-based filtering systems calculate similarity between item profiles based on the item's attributes using various similarity measures. The algorithms are constantly evolving, introducing new similarity estimation methods combining different types of data and classifications such as network and semantic analysis (Son and Kim, 2017). The aim of these methods is to refine the recommender patterns based on large volumes of data related to user preferences. However, as stated above, the evaluation of similarity between a brand and an influencer doesn't involve user preferences. Thus, in this case, even standard similarity measures such as variations of Jaccard similarity will provide accurate recommendations (Bag et al., 2019).

Jaccard Similarity is a similarity measure that is often used in content-based filtering to quantify the similarity between two items or between a user's profile and an item. It is defined as the size of the intersection of the sets of features of two items or a user's profile and an item divided by the size of the union of the sets. In other words, Jaccard Similarity calculates the similarity between two sets by counting the number of items that are common between the two sets and dividing it by the number of items in either set:

$$J(A, B) = |A \cap B| / |A \cup B|$$

where A and B are the sets of features for two items or a user's profile and an item, and |S| represents the number of elements in set S (Koeneman and Cavanaugh, 2022).

Jaccard similarity ranges from 0 to 1, with 1 indicating that the sets are identical and 0 indicating that they have no common features. This makes it a useful measure for comparing the similarity between two items or between a user's profile and an item in a content-based recommender system. Generally, once the similarity between items or users is computed, the recommendation system can use this information to recommend items to users based on the level of similarity. For example, if a similarity is found between a brand and an influencer,

the recommender system can also offer other online creators with a similar attribute evaluation.

However, in order to design a recommender system for brand-influencer collaborations, the main task lies in defining the concept of similarity. Such a system would require a set of criteria with quantifiable ratings that would allow the calculation of Jaccard similarity based on which the matching of the brand and creators would be done. Drawing from the previous research on para-social relationships and celebrity endorsement models, it appears that influencer-follower similarity and influencer-brand congruence can be associated with the perception of the influencer and brand personalities. Therefore, in this article, we develop an influencer-brand similarity measurement scale allowing the comparison of both the brand and the influencer.

#### **4 Methods and Data Collection**

For the initial item selection we used Patrick Gunkel's list of 638 personality traits (Stipp, 1988). Gunkel created this list using the principles of ideonomy. Ideonomy was firstly introduced by Buckminster Fuller, American polymath, in the 1940 to describe his own interdisciplinary approach to understanding the world (Fuller and Snyder, 2010). This list consists of 234 positive traits (37%), 112 neutral traits (18%) and 292 negative traits (46%). All of these items were rated by 82 undergraduate students during a pretest stage. The pretesting consisted in three rounds, one for positive, negative and neutral traits (evaluated by the same group of students), during which the students were asked to rate the traits ability to describe both brand and an influencer. Items were rated on a 7 point scale. We set 5 as a cut-off and online items rated 5 - descriptive, 6 - very descriptive and 7 - extremely descriptive were chosen for next evaluation. This first pretest round led to a reduction from 638 personality traits to 148 negative, 70 neutral and 135 positive traits. In addition to the ratings, the students were asked to write down brands they know that frequently collaborate with influencers on YouTube.

Subsequently, still during the pretest stage, the reduced number of traits was rated by a focus group assembled from 43 employees from three marketing companies. Similarly to the first rating round, we opted for 5 as the cut-off point leading to another reduction on 229 traits. During the focus group, some of the participants suggested that many of the traits are rather synonymous and thus redundant. Therefore, synonyms were eliminated by cross comparison with Thesaurus.com (Dictionary.com, 2022). After this last manipulation, we reached a final count of 102 personality traits.

After the pretest stage, once the list of characteristics was completed, it was necessary to choose the brands and influencers for the scale development. The brands were selected from the list made by the students during the pretest stage based on two criteria: frequency and repetitiveness of the cooperation. Only the brands that previously successfully cooperated with multiple creators on YouTube, while with some of them repeatedly, were selected for evaluation. Our final array of brands thus included ThredUp (online thrift store), Function of Beauty (customized beauty products), Audible (audio book store) and Skillshare (platform with online courses). For each of the brands, we then identified 9 YouTube-based influencers while trying to diversify the sample by including creators with varying numbers of followers ranging from thousands to millions. The list of influencers along with their YouTube channel statistics obtained with online applications Thoughleaders and Socialblade is described in Table 2 (ThoughLeaders, 2022; Socialblade, 2022).

Table 2 YouTube channel statistics

| Brand              | Influencer's Channel         | Number of Followers | Number of Video Views | Number of Uploads | Channel Type  |
|--------------------|------------------------------|---------------------|-----------------------|-------------------|---------------|
| ThredUp            | Rachel Maksy                 | 998K                | 92,747,629            | 318               | Entertainment |
|                    | Morgan Donner                | 335K                | 25,318,859            | 92                | How to        |
|                    | Mia Maples                   | 3.29M               | 541,059,708           | 591               | Entertainment |
|                    | Emma Chamberlain             | 11.6M               | 1,519,386,852         | 282               | Entertainment |
|                    | Ellen Fisher                 | 728K                | 99,975,596            | 265               | Film          |
|                    | Carrie Dayton                | 425K                | 57,798,292            | 447               | People        |
|                    | Hannah Harrell               | 103K                | 1,751,830             | 20                | People        |
|                    | Hannah - The Outfit Repeater | 17.7K               | 2,880,133             | 282               | How to        |
|                    | THE WADS                     | 339K                | 89,971,914            | 1,254             | How to        |
| Audible            | Doctor Mike                  | 9.77M               | 1,791,993,004         | 511               | Education     |
|                    | Dong ASMR                    | 525K                | 92,137,953            | 325               | Entertainment |
|                    | Jazza                        | 6.01M               | 1,203,165,064         | 1,289             | Entertainment |
|                    | Sailing La Vagabonde         | 1.76M               | 359,885,646           | 490               | Travel        |
|                    | Danielle Kirsty              | 603K                | 67,304,232            | 205               | How to        |
|                    | KianAndJe                    | 3.83M               | 812,712,627           | 489               | Entertainment |
|                    | Strange Parts                | 1.84M               | 120,085,770           | 67                | Technology    |
|                    | Eric Rosen                   | 568K                | 142,058,641           | 793               | Games         |
|                    | Gibi ASMR                    | 4.3M                | 1,600,719,250         | 937               | Entertainment |
| Skillshare         | TomKat Stitchery             | 27.7K               | 4,178,476             | 690               | How to        |
|                    | Irene Rudnyk                 | 816K                | 63,691,079            | 311               | Education     |
|                    | REGRETYSS                    | 87.2K               | 7,722,602             | 46                | People        |
|                    | Katnipp                      | 170K                | 13,323,339            | 291               | People        |
|                    | aliyah simone                | 645K                | 38,952,178            | 501               | People        |
|                    | Plant Based Bride            | 137K                | 12,796,306            | 419               | How to        |
|                    | Primink                      | 1.04M               | 102,724,101           | 50                | Entertainment |
|                    | The Bite Shot                | 314K                | 11,380,935            | 196               | Education     |
|                    | Jenna Hong                   | 15.4K               | 1,383,648             | 227               | People        |
| Function of Beauty | Bernadette Banner            | 1.31M               | 98,151,070            | 123               | Education     |
|                    | Glam&Gore                    | 3.95M               | 520,678,365           | 258               | Entertainment |
|                    | Kenna                        | 214K                | 7,403,983             | 112               | How to        |
|                    | Dr Dray                      | 1.63M               | 264,233,211           | 2,304             | How to        |
|                    | Anjeez                       | 26.5K               | 2,733,859             | 170               | How to        |



|  |                |       |            |     |        |
|--|----------------|-------|------------|-----|--------|
|  | Hazel Goddesss | 231K  | 25,095,958 | 698 | How to |
|  | Abbey Yung     | 270K  | 27,438,916 | 418 | How to |
|  | Abby Cox       | 246K  | 17,512,014 | 114 | How to |
|  | Bekah & Co.    | 6.29K | 1,077,437  | 111 | People |

Using the online tool at Similarweb.com (2022), we determined the demographic characteristics and the general interests of the target groups for each of the selected brands. Information related to the brands' audiences is displayed in Table 3.

Table 3 Brand's target group demographics and website performance

| Values per Three Months (April - June) | ThredUp                                                                                             | Audible                                                                                                   | Skillshare                                                                                                  | Function of Beauty                                                                                               |
|----------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Total Visits                           | 12.0M                                                                                               | 23.9M                                                                                                     | 8.2M                                                                                                        | 1.4M                                                                                                             |
| Average Visit Duration                 | 7:30                                                                                                | 3:08                                                                                                      | 3:49                                                                                                        | 3:21                                                                                                             |
| Bounce Rate %                          | 37.70%                                                                                              | 50.55%                                                                                                    | 46.24%                                                                                                      | 46.67%                                                                                                           |
| Pages per Visit                        | 9.55                                                                                                | 4.22                                                                                                      | 3.67                                                                                                        | 4.21                                                                                                             |
| Gender Distribution                    | Female 75.29%, Male 24.71%                                                                          | Female 46.57%, Male 53.43%                                                                                | Female 44.12%, Male 55.88%                                                                                  | Female 65.97%, Male 34.03%                                                                                       |
| Age Distribution                       | 18 - 24 22.04%<br>25 - 34 27.38%<br>35 - 44 15.63%<br>45 - 54 13.06%<br>55 - 64 12.71%<br>65+ 9.18% | 18 - 24 20.00%<br>25 - 34 27.77%<br>35 - 44 18.61%<br>45 - 54 14.49%<br>55 - 64 11.50%<br>65+ 7.63%       | 18 - 24 31.69%<br>25 - 34 34.34%<br>35 - 44 16.03%<br>45 - 54 9.24%<br>55 - 64 5.29%<br>65+ 3.41%           | 18 - 24 30.30%<br>25 - 34 28.15%<br>35 - 44 16.88%<br>45 - 54 10.96%<br>55 - 64 8.28%<br>65+ 5.43%               |
| Audience Interests                     | Fashion and Apparel, News & Media Publishers, Other Computers Electronics and Technology, Beauty    | News & Media Publishers, Other Computers Electronics and Technology, Video Games Consoles and Accessories | Other Computers Electronics and Technology, Programming and Developer Software, Graphics Multimedia and Web | Beauty and Cosmetics, News & Media Publishers, Social Media Networks, Other Computers Electronics and Technology |

|  |  |  |        |  |
|--|--|--|--------|--|
|  |  |  | Design |  |
|--|--|--|--------|--|

Based on the websites demographics of the four brands, we created four testers groups reflecting the characteristics. Testers aged 18-24 years were recruited from the poll of university students. The remaining respondents groups were targeted via Facebook and Instagram ads through which it was possible to reach the needed number of respondents reflecting the needed demographic distribution. The respondents were offered a free online course on Digital photography and Photoshop in exchange for 9 influencer ratings and 1 brand rating. The ratings were done on a 7 point Likert scale. Subsequently, once an exploratory factor analysis (EFA) of the ratings was completed, a confirmatory factor analysis (CFA) was conducted on a confirmatory set of respondents. This new group of respondents was again targeted with Facebook ads setted, thanks to the use of Meta pixel and website cookies, to avoid previous respondents. The confirmatory analysis was conducted for Thred-up since it was the easiest target group of respondents to rectruite during the first round. The characteristics of the respondent groups are summarized in Table 4. Overall, ThredUp and related influencers were evaluated by 49 individuals, Audible by 55, Skillshare by 48, and Function of Beauty by 52. The confirmatory group consisted of 56 respondents.

Table 4 Characteristics of the respondent groups.

|                    | Gender Distribution of the Respondents   | Age Distribution of the Respondents                                                                                            | Total Number of the Respondents |
|--------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| ThredUp            | Female: 36 (73.47%)<br>Male: 13 (26.53%) | 18 - 24 11 (22.45%)<br>25 - 34 14 (28.57%)<br>35 - 44 7 (14.30%)<br>45 - 54 6 (12.24%)<br>55 - 64 5 (10.20%)<br>65+ 6 (12.24%) | 49                              |
| Audible            | Female: 26 (47,27%)<br>Male: 29 (52,73%) | 18 - 24 10 (18.18%)<br>25 - 34 16 (29.09%)<br>35 - 44 11 (20.00%)<br>45 - 54 8 (14.56%)<br>55 - 64 6 (10.90%)<br>65+ 4 (7.27%) | 55                              |
| Skillshare         | Female: 22 (45,83%)<br>Male: 26 (54,17%) | 18 - 24 15 (31.25%)<br>25 - 34 17 (35.42%)<br>35 - 44 8 (16.67%)<br>45 - 54 5 (10.41%)<br>55 - 64 2 (4.17%)<br>65+ 1 (2.08%)   | 48                              |
| Function of Beauty | Female: 34 (65,38%)<br>Male: 18 (34,62%) | 18 - 24 16 (30.77%)<br>25 - 34 14 (26.92%)<br>35 - 44 10 (19.23%)<br>45 - 54 6 (11.54%)<br>55 - 64 5 (9.62%)                   | 52                              |

|                                       |                                                |                                                                                                                              |    |
|---------------------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----|
|                                       |                                                | 65+ 1 (1.92%)                                                                                                                |    |
| ThredUp<br>(Confirmatory<br>Data Set) | Female: 42<br>(75.00%)<br>Male: 14<br>(25.00%) | 18 - 24 13 (23.21%)<br>25 - 34 18 (32.14%)<br>35 - 44 8 (14.29%)<br>45 - 54 7 (12.5%)<br>55 - 64 6 (10.71%)<br>65+ 4 (7.15%) | 56 |

## 5 RESULTS

For each of the brands we conducted an exploratory factor analysis using the principal components method of extraction without specifying the number of factors. This method generated 30 factors explaining 75% to 76% of total variance. The Kaiser-Meyer-Olkin Measure of Sampling Adequacy reached higher values than 0.8 for three of the brand-focused EFAs indicating a sampling adequacy. However, for Skillshare, the matrix was identified as not positive definite (Table 5). For the confirmatory factor analysis we used only factors with coefficients higher than 0.7 and at least 3 factor loadings per factor. The CFA performed on the additional ThredUp data set showed that the 8 factors explained 83% of total variance with 0.9 KMO.

Table 5 Exploratory and confirmatory analysis results

|                           | Total<br>Variance<br>explained % | Number of<br>Extracted<br>Components | Kaiser-Meyer-<br>Olkin Measure of<br>Sampling<br>Adequacy | Bartlett's Test of<br>Sphericity<br>Approx.     |
|---------------------------|----------------------------------|--------------------------------------|-----------------------------------------------------------|-------------------------------------------------|
| EFA Audible               | 75.194                           | 30                                   | .880                                                      | Chi-Square<br>51857.289<br>df 5151<br>Sig. .000 |
| EFA Function<br>of Beauty | 75.215                           | 30                                   | .867                                                      | Chi-Square<br>46838.027<br>df 5151<br>Sig. .000 |
| EFA Skillshare            | 75.205                           | 30                                   | Matrix not<br>positive definite                           | Matrix not positive<br>definite                 |
| EFA Thredup               | 76.004                           | 30                                   | .871                                                      | Chi-Square<br>48233.764<br>df 5151<br>Sig. .000 |
| CFA Thredup               | 83.084                           | 8                                    | .901                                                      | Chi-Square<br>40780.190<br>df 1431<br>Sig. .000 |

The EFA thus led to a reduction from 30 factors to 8 factors and from 102 factor loadings to 54 factor loadings. Individual factor loadings for the EFA and CFA are described in Tables 6 -13. All the factor loadings for the 8 factors we identified, reached 0.8 and higher values. According to our final results, the first factor is represented by 10 factor loadings describing personality traits that are generally associated with a person's likeability. Therefore we call this factor Agreeableness. Similarly to Agreeableness, the second factor consists of 10 personality traits. However, in this case, the characteristics have rather negative connotations. Influencers with these traits may be perceived as controversial. On the other hand, controversial influencers shouldn't be categorized as unsuitable for brand collaborations. The proposed framework builds upon the match-up hypothesis and thus the second factor, Tendentiousness, should be also treated as neutral since the evaluation is based on finding a suitable match that is consistent with the brand's image and values. Controversial creators are very likely to have a good fit with controversial brands.

The following group of 9 factor loadings may be associated with ambitious and determined individuals who strive for achievements and have a set of goals they want to achieve in life. We thus named the third factor Drivenness. Factors 4, Spontaneousness, and 6, Creativeness, are, much like the previous three groups, related to temperament and capacities of the influencer. Conversely, the last remaining factors reflect physical attributes of the influencer. The factors are Styling, Attractiveness and Sophistication.

Table 6 Factor loadings for Agreeableness.

|               | Factor 1: Agreeableness |                        |                |             |             |
|---------------|-------------------------|------------------------|----------------|-------------|-------------|
|               | EFA Audible             | EFA Function of Beauty | EFA Skillshare | EFA Thredup | CFA Thredup |
| Kind          | .914                    | .885                   | .873           | .875        | .859        |
| Loyal         | .912                    | .934                   | .908           | .813        | .921        |
| Genuine       | .910                    | .911                   | .869           | .862        | .969        |
| Earnest       | .898                    | .952                   | .893           | .879        | .960        |
| Transparent   | .893                    | .964                   | .864           | .864        | .903        |
| Humble        | .890                    | .930                   | .849           | .871        | .898        |
| Conscientious | .883                    | .939                   | .875           | .863        | .965        |
| Sympathetic   | .883                    | .991                   | .908           | .933        | .866        |
| Helpful       | .873                    | .935                   | .858           | .861        | .928        |
| Appreciative  | .864                    | .955                   | .858           | .895        | .857        |

Table 7 Factor loadings for Tendentiousness.

|             | Factor 2: Tendentiousness |                        |                |             |             |
|-------------|---------------------------|------------------------|----------------|-------------|-------------|
|             | EFA Audible               | EFA Function of Beauty | EFA Skillshare | EFA Thredup | CFA Thredup |
| Superficial | .891                      | .912                   | .872           | .953        | .915        |
| Artificial  | .888                      | .900                   | .871           | .981        | .972        |
| Selfish     | .886                      | .908                   | .867           | .918        | .927        |

|                 |      |      |      |      |      |
|-----------------|------|------|------|------|------|
| Careless        | .861 | .980 | .917 | .932 | .946 |
| Narcissistic    | .859 | .900 | .887 | .917 | .884 |
| Individualistic | .844 | .887 | .892 | .905 | .842 |
| Disrespectful   | .844 | .929 | .859 | .900 | .958 |
| Bizarre         | .841 | .792 | .859 | .943 | .875 |
| Cynical         | .840 | .852 | .872 | .933 | .869 |
| Impulsive       | .821 | .884 | .859 | .942 | .921 |

Table 8 Factor loadings for Drivenness.

|               | Factor 3: Drivenness |                        |                |             |             |
|---------------|----------------------|------------------------|----------------|-------------|-------------|
|               | EFA Audible          | EFA Function of Beauty | EFA Skillshare | EFA Thredup | CFA Thredup |
| Big-thinking  | .907                 | .934                   | .805           | .901        | .893        |
| Determined    | .887                 | .909                   | .783           | .855        | .941        |
| Perfectionist | .861                 | .883                   | .735           | .876        | .860        |
| Competitive   | .843                 | .884                   | .726           | .858        | .905        |
| Extravagant   | .840                 | .921                   | .766           | .848        | .830        |
| Progressive   | .840                 | .926                   | .733           | .849        | .879        |
| Daring        | .834                 | .875                   | .755           | .820        | .738        |
| Ambitious     | .830                 | .918                   | .764           | .866        | .940        |
| Outspoken     | .789                 | .920                   | .790           | .861        | .893        |

Table 9 Factor loadings for Spontaneousness

|              | Factor 4: Spontaneousness |                        |                |             |             |
|--------------|---------------------------|------------------------|----------------|-------------|-------------|
|              | EFA Audible               | EFA Function of Beauty | EFA Skillshare | EFA Thredup | CFA Thredup |
| Playful      | .853                      | .987                   | .857           | .802        | .908        |
| Adventurous  | .834                      | .953                   | .792           | .784        | .966        |
| Curious      | .833                      | .951                   | .837           | .789        | .947        |
| Energetic    | .829                      | .955                   | .832           | .790        | .908        |
| Freethinking | .828                      | .926                   | .838           | .724        | .864        |
| Amusing      | .815                      | .956                   | .822           | .777        | .956        |
| Messy        | .811                      | .927                   | .838           | .748        | .875        |

Table 10 Factor loadings for Styling

|  | Factor 5: Styling |                        |                |             |             |
|--|-------------------|------------------------|----------------|-------------|-------------|
|  | EFA Audible       | EFA Function of Beauty | EFA Skillshare | EFA Thredup | CFA Thredup |

|              |      |      |      |      |      |
|--------------|------|------|------|------|------|
| Athletic     | .830 | .905 | .807 | .785 | .854 |
| Businesslike | .812 | .949 | .769 | .733 | .834 |
| Pure         | .800 | .912 | .796 | .801 | .828 |
| Romantic     | .798 | .925 | .776 | .796 | .926 |
| Escapist     | .769 | .928 | .718 | .776 | .950 |
| Rustic       | .761 | .925 | .774 | .828 | .889 |
| Idealistic   | .703 | .964 | .769 | .809 | .949 |

Table 11 Factor loadings for Creativeness

|             | Factor 6: Creativeness |                        |                |             |             |
|-------------|------------------------|------------------------|----------------|-------------|-------------|
|             | EFA Audible            | EFA Function of Beauty | EFA Skillshare | EFA Thredup | CFA Thredup |
| Imaginative | .879                   | .948                   | .883           | .877        | .946        |
| Creative    | .867                   | .983                   | .792           | .860        | .952        |
| Resourceful | .861                   | .947                   | .845           | .870        | .954        |
| Innovative  | .839                   | .949                   | .813           | .856        | .881        |

Table 12 Factor loadings for Attractiveness

|             | Factor 7: Attractiveness |                        |                |             |             |
|-------------|--------------------------|------------------------|----------------|-------------|-------------|
|             | EFA Audible              | EFA Function of Beauty | EFA Skillshare | EFA Thredup | CFA Thredup |
| Charismatic | .908                     | .975                   | .905           | .904        | .964        |
| Attractive  | .889                     | .938                   | .878           | .885        | .930        |
| Intelligent | .887                     | .925                   | .876           | .866        | .920        |
| Trendy      | .885                     | .909                   | .884           | .876        | .925        |

Table 13 Factor loadings for Sophistication

|               | Factor 8: Sophistication |                        |                |             |             |
|---------------|--------------------------|------------------------|----------------|-------------|-------------|
|               | EFA Audible              | EFA Function of Beauty | EFA Skillshare | EFA Thredup | CFA Thredup |
| Sophisticated | .897                     | .979                   | .901           | .947        | .910        |
| Tasteful      | .874                     | .971                   | .877           | .932        | .886        |
| Elegant       | .862                     | .969                   | .873           | .920        | .910        |

Finally, once the individual factors were established, the fit between influencers and Thredup in the confirmatory data set was calculated using the Jaccard similarity index for each pair of influencer-brand ratings. Since Jaccard index is calculated as a binary similarity all respondent ratings for the observed attributes were binary coded. Ratings ranging from 1 to 4 were considered as 0 values and ratings above 4 as 1 values. The Jaccard similarity

coefficients for the confirmatory Thredup data set are displayed in Table 14. Since the influencer channels were selected based on repeated cooperation with the brands, a good influencer-brand fit was expected. The Jaccard similarity coefficients in Table 14 reflect this high degree of congruence between the studied channels and Thredup.

Table 14 Jaccard similarity coefficients per respondent for the Thredup confirmatory data set.

|               | Rachel<br>Maksy | Carrie<br>Dayton | Ellen<br>Fisher | Emma<br>Chamberlain | Hannah -<br>The Outfit<br>Repeater | Hannah<br>Harrell | Mia<br>Maples | Morgan<br>Donner | The<br>WAD<br>S |
|---------------|-----------------|------------------|-----------------|---------------------|------------------------------------|-------------------|---------------|------------------|-----------------|
| Respondent 1  | 1.000           | .970             | .970            | .970                | .970                               | .939              | .939          | .970             | .914            |
| Respondent 2  | .939            | .970             | .970            | .970                | .882                               | .970              | .909          | .939             | .829            |
| Respondent 3  | .970            | .970             | .970            | .970                | .914                               | .970              | .939          | .970             | .879            |
| Respondent 4  | .970            | .879             | .939            | .879                | .938                               | .970              | .939          | .939             | .879            |
| Respondent 5  | .879            | .970             | .970            | .970                | .914                               | .970              | .970          | .909             | .914            |
| Respondent 6  | .970            | .970             | .970            | .970                | .914                               | .970              | .909          | .970             | .914            |
| Respondent 7  | .939            | .970             | .909            | .879                | .882                               | .970              | .909          | .970             | .970            |
| Respondent 8  | .970            | .970             | .909            | .970                | .938                               | .970              | .970          | .970             | .879            |
| Respondent 9  | .879            | .970             | .970            | .970                | .970                               | .970              | .909          | .909             | .970            |
| Respondent 10 | .970            | .970             | .970            | .879                | .970                               | .970              | .970          | .909             | .879            |
| Respondent 11 | .909            | .970             | .909            | .970                | .970                               | .970              | .909          | .909             | .970            |
| Respondent 12 | .970            | .970             | .939            | .970                | .941                               | .970              | .939          | .970             | .879            |
| Respondent 13 | .970            | .970             | .970            | .970                | .970                               | .939              | .970          | .970             | .914            |
| Respondent 14 | .909            | .970             | .939            | .970                | .941                               | .879              | .970          | .970             | .914            |
| Respondent 15 | .939            | .970             | .970            | .970                | .970                               | .909              | .970          | .970             | .970            |
| Respondent 16 | .909            | .970             | .909            | .879                | .941                               | .879              | .909          | .970             | .914            |
| Respondent 17 | .970            | .970             | .939            | .970                | .941                               | .970              | .970          | .970             | .914            |
| Respondent 18 | .909            | .970             | .970            | .879                | .970                               | .970              | .909          | .879             | .970            |
| Respondent 19 | .970            | .970             | .970            | .970                | .939                               | .970              | .970          | .970             | .914            |

|               |      |      |      |      |      |      |      |      |      |
|---------------|------|------|------|------|------|------|------|------|------|
| Respondent 20 | .970 | .970 | .939 | .970 | .912 | .970 | .909 | .970 | .914 |
| Respondent 21 | .970 | .970 | .970 | .970 | .969 | .970 | .909 | .879 | .970 |
| Respondent 22 | .970 | .970 | .970 | .970 | .939 | .879 | .970 | .879 | .970 |
| Respondent 23 | .970 | .970 | .970 | .970 | .970 | .970 | .909 | .970 | .879 |
| Respondent 24 | .970 | .970 | .909 | .970 | .941 | .970 | .970 | .970 | .970 |
| Respondent 25 | .970 | .970 | .970 | .970 | .970 | .970 | .970 | .970 | .970 |
| Respondent 26 | .970 | .970 | .970 | .970 | .939 | .970 | .970 | .970 | .829 |
| Respondent 27 | .970 | .970 | .970 | .970 | .970 | .970 | .970 | .970 | .914 |
| Respondent 28 | .970 | .970 | .909 | .970 | .969 | .970 | .970 | .970 | .829 |
| Respondent 29 | .970 | .970 | .970 | .879 | .941 | .970 | .879 | .970 | .879 |
| Respondent 30 | .879 | .970 | .909 | .879 | .970 | .914 | .970 | .970 | .914 |
| Respondent 31 | .879 | .970 | .970 | .970 | .969 | .970 | .970 | .970 | .970 |
| Respondent 32 | .879 | .970 | .970 | .970 | .970 | .970 | .970 | .970 | .970 |
| Respondent 33 | .970 | .970 | .879 | .970 | .914 | .970 | .970 | .970 | .914 |
| Respondent 34 | .970 | .970 | .879 | .970 | .970 | .970 | .970 | .970 | .879 |
| Respondent 35 | .970 | .970 | .970 | .970 | .970 | .970 | .970 | .879 | .970 |
| Respondent 36 | .970 | .879 | .848 | .970 | .970 | .970 | .970 | .970 | .879 |
| Respondent 37 | .939 | .879 | .970 | .970 | .938 | .914 | .970 | .970 | .970 |
| Respondent 38 | .970 | .879 | .879 | .970 | .970 | .970 | .970 | .939 | .970 |
| Respondent 39 | .909 | .970 | .970 | .970 | .970 | .879 | .970 | .970 | .970 |
| Respondent 40 | .909 | .970 | .879 | .879 | .914 | .879 | .970 | .909 | .879 |
| Respondent 41 | .970 | .879 | .970 | .970 | .914 | .970 | .970 | .939 | .970 |
| Respondent 42 | .970 | .970 | .879 | .939 | .970 | .914 | .970 | .939 | .879 |
| Respondent 43 | .970 | .970 | .970 | .909 | .882 | .857 | .970 | .970 | .970 |
| Respondent 44 | .970 | .879 | .970 | .939 | .939 | .909 | .970 | .970 | .970 |



|               |      |      |      |      |      |      |      |      |      |
|---------------|------|------|------|------|------|------|------|------|------|
| Respondent 45 | .970 | .970 | .970 | .970 | .939 | .914 | .970 | .909 | .970 |
| Respondent 46 | .970 | .970 | .879 | .939 | .914 | .970 | .970 | .970 | .970 |
| Respondent 47 | .970 | .970 | .970 | .939 | .970 | .939 | .970 | .970 | .879 |
| Respondent 48 | .970 | .970 | .939 | .970 | .969 | .909 | .970 | .970 | .970 |
| Respondent 49 | .909 | .879 | .970 | .970 | .912 | .886 | .970 | .970 | .970 |
| Respondent 50 | .970 | .970 | .970 | .970 | .970 | .970 | .970 | .939 | .970 |
| Respondent 51 | .970 | .879 | .879 | .879 | .912 | .970 | .879 | .939 | .970 |
| Respondent 52 | .970 | .970 | .970 | .939 | .914 | .914 | .879 | .939 | .879 |
| Respondent 53 | .939 | .970 | .970 | .970 | .912 | .970 | .970 | .939 | .970 |
| Respondent 54 | .970 | .970 | .970 | .970 | .970 | .939 | .970 | .939 | .970 |
| Respondent 55 | .970 | .970 | .970 | .970 | .970 | .970 | .970 | .879 | .970 |
| Respondent 56 | .939 | .909 | .970 | .970 | .970 | .970 | .879 | .970 | .970 |

## 5 Discussion and implications

### 5.2. Theoretical implications

The theoretical contributions of this article are threefold. Firstly, this article uses perceived similarity as an universal measure conditioning all other dimensions of influencer-follower para-social relationships and both positive and negative activations induced by the influencer, brand and product fit. Perceived similarity is in this case treated as the final outcome of subjective processing of the influencer and brand personalities including attitude homophily, values and attractiveness. The perception of both the influencer and the brand is manifested by the description of their personalities. Unlike previously used scales (Hovland and Weiss, 1951; McGuire's, 1985; McCracken, 1989; Kamins, 1990), the proposed framework is tailored for the influencer-brand matching and as such it assigns more humanized aspects to branding. The proposed framework perceives influencers as brands with strong para-social relationships and in order to establish similarity between them and regular brands, it requires extension of the existing brand personality scales.

The second theoretical contribution lies in the overlap of the traditional celebrity endorsement concepts with information science. The current trends in influencer marketing indicate a need for brand and influencer congruence measurement frameworks that would be easy to use and implement in an semi-automatic manner as part of online recommender systems and applications. The urge to comply with the requirements of recommender systems implies that a set of quantifiable parameters should be selected and that these parameters should reflect both the influencer and the brand to allow the similarity calculation. In addition, this quantification should be done in a manageable manner for a large number of online creators and brands. Unlike the traditional evaluation approaches (Hovland and Weiss, 1951;

McGuire's, 1985; McCracken, 1989; Kamins, 1990), this information system-based approach demands the simplification of the measurement process of the influencer and brand congruence to minimize the efforts of the potential users.

The third contribution of this paper consists in the data collection methodology combining website analytics, performance metrics from the online creators channels and targeted advertising to recruit respondents reflecting the composition of the brands and influencers target audiences. While in this study used only as a means to the measurement scale development, this method could be implemented as part of the influencer brand recommender systems. Automatic comparison of the brand's website traffic data with the creators channel data would help to create the preliminary congruence score based on the audience overlap. Subsequently, this matching could be sophisticated by the 8 factor framework identified by the EFA and CFA performed above. This article thus lays down a foundation for new interdisciplinary research in the area of influencer brand congruence estimation.

### *5.3. Managerial implications*

For influencer endorsements to be effective, an alignment among three intrinsic contributors is necessary: the influencer, the consumer/follower and the brand (Sokolova and Kefi, 2020). Each of these contributors has their own personality and perception of self (Donvito et al., 2020). A strong alignment between consumers, influencers and brands generates more positive attitudes towards the product and higher purchase and recommendation intentions, ensuring optimal returns from influencer marketing campaigns (Belanche et al., 2021). However, due to the complexity of the para-social relationships between influencers and their followers, measuring the level of congruence between the brand and the influencer is a daunting task. Existing models consider complex abstract concepts such as credibility or attitude homophily. These techniques are very often based on self-evaluation where the potential consumers answer questions regarding their self-concept, values and attitudes. These are subsequently compared with their evaluation of the celebrities and influencers (Torres et al., 2019). Unfortunately, this approach requires advanced psychometric scales and assessment techniques.

While psychometric scales and assessments are highly recommended for research purposes, they are rather inaccessible and impractical for practitioners. The method and measurement factors demonstrated in this paper represent an easy to use framework. The influencer-brand congruence framework is divided into four steps: identification of the brands target audience, selection of related influencers, evaluation of the brand and the selected influencers using the 8 factors, and computation of the Jaccard similarity as the congruence indicator. All the four steps reflect the mechanisms used by recommender systems and thus can be semi-automatized with online applications. Although the actual software development wasn't the main goal of this article, the applicability of individual steps was demonstrated with external applications. To identify the audiences of the four brands, we used a website analytical application Similarweb.com (2022) that provided not only basic demographic information, but also general interests of the users who interacted with the brands' websites. Similarly, the 8 factor evaluation framework including the Jaccard similarity was turned into a simple online questionnaire with an automatic score calculator. All these qualities underline the applicational potential of this research on influencer-brand congruence in the context of emerging influencer-brand collaboration platforms.

## **6 Limitations and Directions for Future Research**

Framework proposed in this article aims at facilitating brands and influencers finding endorsement collaborations that represent the best fit with their brand personality and products. Considering the possible integration with emerging recommender systems for brands and influencers, the match-up hypothesis was chosen as the main theoretical basis of this research and thus the influencer-brand congruence framework is suffering from the same shortcomings as its foundation. Similarly to the match-up hypothesis, the framework assumes that perceived attractiveness, credibility and expertise of the influencer are the results of the congruence between consumers' perceived self-concept and the attitude homophily with the brand and the influencer. However, the framework does not directly measure this relationship. The consumer is omitted from the ratings based on the prerequisite that the followers' self-concept is represented by the influencer persona. Therefore there is an expectation that, if the brand finds an influencer with high level of congruence, the followers will establish the same attitude homophily with the endorsed brand as with the influencer.

While previous research on celebrity endorsement and current research on influencer endorsement suggest that the relationship between influencers and their followers may positively affect the endorsed brand, it was not directly measured in the context of the congruence framework presented in this paper. Although the exploratory and confirmatory factor analysis were carried out with ratings of influencers and brands provided by potential consumers, future studies might consider extending the validity of this framework by testing it on actual followers of the selected influencers. However, such a study would probably require an extensive effort for data collection since the names of followers are either hidden (YouTube) or the personal accounts are difficult to directly address (Facebook, Instagram). The recruitment of the followers would thus require posting offers in the comments sections of the influencers' channels which might be considered as unsolicited content.

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